



Politehnica University of Timisoara



Web Programming

C1. Overview. Web Foundation



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- Exam – Technical report and presentation
- Final mark: 50% Exam, 50% Lab
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Motivation

- Web is everywhere
 - From personal computers in the home and workplace to mobile devices like Cars, Tablets or Smartphones
 - 01.2025: 20 to 60 billion Web pages (<http://www.worldwidewebsize.com/>)
- Web programming is a leading topic, connected with IoT, Cloud, AI etc.
- Web becomes a programming platform



Course goals

- Discuss the state of the art in Web programming techniques
- Study some of the latest development frameworks and tools
- Understand the emergent technologies, such as the Web 3.0 or AI-assisted Web development



Contents

- Overview, Web foundation
- Client-Side Programming
- Server-Side Programming
- Ajax. Web App Frameworks
- Rich Internet Applications
- Web Multimedia Applications
- Emergent Technologies
- AI-assisted Web Development



Overview. Web foundation

■ Web Foundation

- HTML



- XML



- XHTML

- CSS



- RDF(S)



- OWL

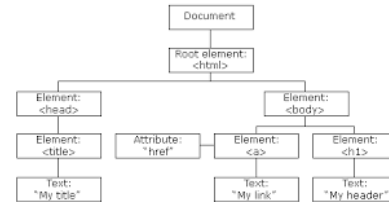




Client-Side Programming

■ Client-Side Scripting

- Scripting languages. JavaScript
- DOM
- Client-page interaction
- JQuery. Bootstrap
- React
- VueJS
- Flutter Web/Dart

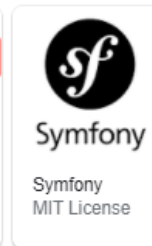




Server-Side Programming

■ Server-Side Scripting

- PHP.ASP. JSP
- NodeJS+Express
- Java/Kotlin Spark
- WAFs
 - Django. Flask
 - Symfony
 - CodeIgniter
 - Laravel





Ajax. Web App Frameworks

- Ajax

- XMLHttpRequest
- Managing interactions
- Ajax libraries



- Ajax based Web App Frameworks

- Dojo Toolkit
- Angular
- NextJS





Rich Internet Applications

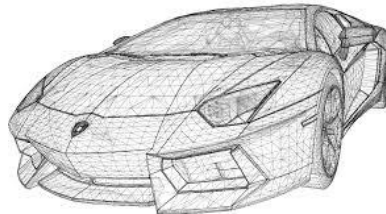
- Rich Internet Applications (RIA)
 - Google APIs ([Google Maps API](#), [Google Visualization API](#), [Google Picker API](#))
 - Web services / Web mashups
 - Java Spring
 - Node.js+React+Redux
 - Performance - Web Assembly
 - Security and protection
 - Testing ([Selenium](#), [Windmill](#))





Web multimedia applications

- Web multimedia applications
 - SMIL – Synchronized Multimedia Integration Language
 - jsPsych
 - Web CMS: Joomla, Wordpress
 - Scalable Vector Graphics (SVG)





Emergent Technologies

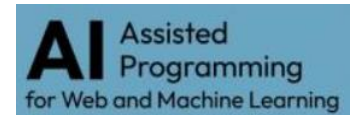
- Social Web (web 2.0) -> Semantic Web (3.0)
 - *Social Web: social media, user-generated content*
 - *Semantic Web: a group of methods and technologies to allow machines to understand the meaning - or "semantics" - of information on the World Wide Web*
 - Semantic Web Data and Ontologies
 - Semantic Web Services
 - NLP and Data Mining on Semantic Web
- WebIOT





AI-assisted Web Development

- AI-assisted development
 - improve productivity by writing practical and relevant prompts to build web apps
 - improve the maintainability and performance of existing code
 - GitHub Copilot/ChatGPT prompts for coding
 - ML models integration for Web AI applications





AI in Web development. Examples (I)

- Intelligent Chatbots (e.g., personalized customer support)
 - Plug-ins from platform like Intercom, Birdeye, Drift
- Smart Content Generation
 - Natural Language Generation (NLG) systems can write news article or product description (e.g., ChatGPT, Narrato, Jasper)
- Advanced Data Analytics
 - It help web developers improve website performance, enhance conversion rates, or tailor marketing strategies Outset.ai, Deepsheet, Jeeves)





AI in Web development. Examples (II)

- Personalized User Experience



- AI analyze browsing history, preferences, and demographics. By using ML, sites can dynamically adjust their content, layout, and recommendations (UIzard, Attention Insight, Khroma)



- Automated Testing and Bug Detection

- As for general software dev (TestCraft, Applitools, Functionize, Mabl)



- Design Assistance



- AI can generate design recommendations, templates, or page layouts (Fontjoy, Zyro, 10Web, Designs.AI)

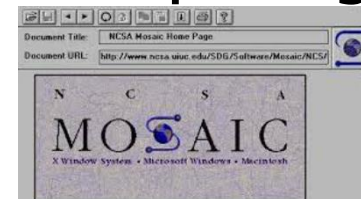


WEB Foundation



Internet and Web

- '80 "*World Wide Web*" - Tim Berners-Lee and Robert Cailliau at CERN
- Most of the internet protocols was adopted in early '90 (TCP-IP, FTP, HTTP ...)
- 1992 National Center for Supercomputing Applications – NCSA Mosaic
- 1994 – the WWW consortium (MIT, INRIA, DARPA, EU Commission)





HTML

- Originally developed by Tim Berners-Lee in 1990 at CERN (Conseil Européen pour la Recherche Nucléaire)
- Supported by the NCSA Mosaic browser
- Should work well across different browsers and platforms
- W3c standard (HTML5.2 - 2017, XHTML)
- HTML 5 - live by WHATWG (Web Hypertext Application Technology Working Group) Apple, Mozilla, Google, Microsoft



HTML Document Structure

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">
<HTML>
<HEAD>
  <TITLE>Title</TITLE>
</HEAD>

<BODY>
<H1>Main Heading</H1>

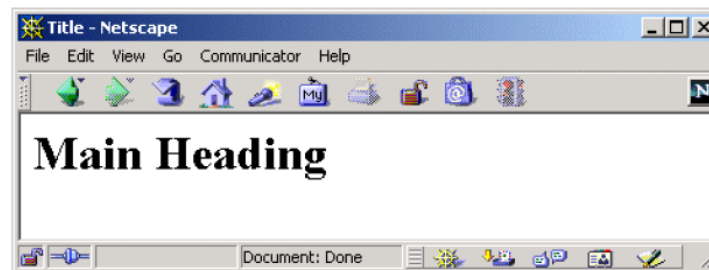
<!-- Rest of page goes here -->

</BODY>
</HTML>
```

← Goes on browser's title bar
May not appear in printouts

← Main heading. Often used as title
Appears in printouts

← HTML comment
Replace with body of
WWW page





HTML elements

- Head: Title, Meta, Bgsound, Style
 - Meta: description, keywords, author, HTTP_EQUIV = "refresh"
- Text: P, BR, HR, H1..H6, DIV, SPAN
- Special chars: < > & "
- List: OL, UL, LI, DL, DT, DD
- Image: IMG, MAP/AREA, PICTURE/SOURCE
- Table: TABLE, TH, TR, TD, COLGROUP
- Link: A



HTML FORMS elements

- `<FORM>` - display an HTML form
 - ACTION - URL of a SSC or CGI program
 - METHOD - how to transfer the data to the server
- `<INPUT>` - creates a data entry field
- `<TEXTAREA>` - a form field for large text
- `<SELECT>` - a selectable list
- `<OPTION>` - an option inside a `<select>` list
- `<BUTTON>` - creates a button
- `<LABEL>` - a text associated with an element



HTML 5

- The latest standard
- Browsers offer partial support for it (e.g. Chrome, Opera, Firefox, Safari, IE)
- Compliance test: <https://html5test.co/>
- Aims
 - Reduce the need for external plugins (like Flash)
 - Better error handling
 - More markup to replace scripting
 - Device independent (e.g. various mobile devices)



HTML 5 – New Features

- New features:
 - **Semantic** elements: <header>, <footer>, <article>, <section>
 - **Controls** elements : number, date, time, calendar, range
 - **Graphic** elements: <svg>, <canvas>
 - **Multimedia** elements : <audio>, <video>.



HTML 5 – New APIs

New APIs:

- HTML Geolocation: used to locate a user's position
- HTML Drag and Drop: any element can be draggable
- HTML Local Storage: web pages can store data locally (replace cookies technology)
- HTML Application Cache: for making an offline version of a web application
- HTML Web Workers: JavaScript running in the background
- HTML SSE: Server-Sent Events to get page updates



HTML 5 – Custom elements

```
<head>
  <title>Creating an HTML Element</title>
  <script>document.createElement("niceText")</script>
  <style>
    niceText {
      display: block;
      color: red;
      font-size: 25px;
    }
  </style>
</head>
<body>
  <h1>New element: &lt;niceText&gt;</h1>
  <niceText>This is an example of nice text!</niceText>
</body>
```



HTML 5 – Media Tags

- `<video>`
 - video content: movie clips (mpeg4, ogg, WebM)
- `<audio>`
 - audio streams: sounds, music (mp3, ogg vorbis, wav)
- `<source>`
 - for multiple formats, defined inside video or audio elements
- `<embed>`
 - for embedded content, such as a plug-in
- `<canvas>`
 - runtime drawings with a script (e.g. javascript)



HTML 5 – Data Storage

- Two new objects for storing data on the client
 - `localStorage` - stores data with no time limit
 - `sessionStorage` - stores data for current session
- Data is not passed-on by every server request but used only when asked for. It is possible to store large amounts of data without affecting the website's performance
- The data is stored in different areas for different websites, and a website can only access data stored by itself
- JavaScript is used to store and access the data.



HTML 5 – Local Storage

- E.g. Page counter

```
<script type="text/javascript">
  if (localStorage.counter)
  {
    localStorage.counter=Number(localStorage.counter)+1;
  }
  else
  {
    localStorage.counter=1;
  }
  document.write("Visitor #"+localStorage.counter);
</script>
```



HTML 5 – New form elements

- Several complex form input elements:

<body>

Select a color: <input type="color" name="myColor" />

Select a Date: <input type="date" name="myDate" />

Choose a number (between 1 and 9):

<input type="number" name="myNumber" min="1" max="9" />

Select from a range:

<input type="range" name="myRange" min="1" max="9" />

</body>



XHTML

- XHTML (eXtensible Hypertext Markup Language) - a family of XML languages developed to extend HTML
- The main goal: to make HTML more flexible and increase interoperability with other data formats through XML mechanisms
- Allows mixing of HTML code with other XML based languages like Scalable Vector Graphics (SVG) or MathML



XHTML versus HTML

- There are several differences between HTML and XHTML:
 - all elements (tags) has to be closed (incl. `<input/>` or `<br/>`)
 - XML is case-sensitive for element and attribute names
 - XML does not allow attribute minimization (ex. `<option selected="selected">`)
 - behavior on parse errors differ (processing could be aborted)
 - support for namespaces
 - differences in JavaScript processing (ex. `document.write()` will not work in XHTML)
 - There are three different XHTML 1.0 corresponding to the three different versions of HTML 4 as: XHTML 1.0 Strict, XHTML 1.0 Transitional and XHTML 1.0 Frameset.



CSS

- CSS - Cascading Style Sheets is a language used to describe the appearance (the look and formatting) of a document written in a markup language (ex. HTML)
- The main objective was to enable the separation of document content from the document presentation
- CSS specifications are maintained by the W3C



CSS Syntax

- Three Ways to Insert CSS
 - External style sheet
 - Internal style sheet
 - Inline style
- Syntax: `selector { propriety: value;... }`
- Selectors
 - Element: `p {color:red;}`
 - Class: `.redText {color:red;}`
 - Id: `#textC {text-align:center;}`
 - Hierarchical: `#bigTable td {text-align:left;}`



CSS Solving Conflicts

```
<html>
<head>
  <style type="text/css">
    .red{
      font-size:15px;
      color:red;
    }
  </style>
  <title>Apple Example</title>
</head>
<body>
  <p class="red" style="color:green">Apple</p>
</body>
</html>
```



CSS Styling Text

color: .. color.. | ..hex.. | ..rgb..
direction: ltr | rtl
letter-spacing: normal | ..px..
word-spacing: normal | ..px..
text-align: left | right | center | justify
text-decoration: none | underline | line-through |
blink
text-indent: ..px.. | %
text-transform: none | capitalize | uppercase |
lowercase
white-space: normal | pre | nowrap



CSS Styling Lists and Tables

`list-style-image: none | url`

`list-style-position: inside | outside`

`list-style-type: disc | circle | lower-greek | ..`

`border: size border-type color;`

`border-collapse: separate | collapse;`

`width: size;`

`height: size;`

`text-align: left | right | center;`

`padding: size;`

`color: color | rgb(r,g,b);`

`background-color: color | rgb(r,g,b);`



CSS - ref

- CSS Tutorial

<http://www.w3schools.com/css/default.asp>

- Free CSS Menu Editor

<http://purecssmenu.com/>



Technologies for the Semantic Web

- Metadata
 - Resources are marked-up with descriptions of their content. Need for a standard language
- Terminologies
 - Provide common vocabularies of a domain, so search engines, agents and users can communicate. No good unless everyone means the same thing
- Ontologies
 - Provide a shared and common understanding of a domain for humans and applications to support information exchange and discovery



OWL: Web Ontology Language

(Courtesy of S. Bechhofer, University of Manchester, UK)

- An ontology language designed for Web
 - A rich collection of operators for forming concept descriptions
 - Is a W3C standard, promoting interoperation and sharing between applications
 - Has been designed to be compatible with existing web standards
- In the Semantic Web, resources should be more accessible to automated processes
 - Via semantic markup
 - Metadata annotations that describe content/function



RDF

- RDF stands for Resource Description Framework
- It is a W3C Recommendation
 - <http://www.w3.org/RDF>
- It is a graphical formalism (+ XML syntax)
 - for representing metadata
 - for describing the semantics of information in a machine accessible way
- Provides a simple data model based on triples



OWL

- A formally specified language as solution to RDF(S) problems
- Based on the DAML+OIL specification
- Well defined RDF/XML serializations
- Formal semantics
 - First Order
 - Relationship with RDF
- Comprehensive test cases for tools/ implementations



OWL Description Logic Semantics

- Model theoretic semantics. An interpretation consists of
 - A domain of discourse (a collection of objects)
 - Functions mapping
 - Classes to sets of objects
 - Properties to sets of pairs of objects
 - Rules describing how to interpret the constructors
- In a DL, a class description is a characterization of the individuals that are members of that class



Query Languages

- SPARQL is a proposed query language for RDF
<http://www.w3.org/TR/rdf-sparql-query/>
- SPARQL defines Protocol, Query Language and Results Format
- Query language is the main part
 - Protocol allows query, no update
 - Variety of results formats supported as XML, JSON (used in Web 2.0 apps) and RDF
- SPARQL for OWL (OWL's standard syntax is RDF)